

KATTUMURI MADHU

FULL-STACK DEVELOPER | AI&ML | PYTHON DEVELOPER

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SUMMARY

Full-Stack Developer with a strong foundation in Python, AI/ML, React, Next.js, Node.js, and MongoDB. Experienced in building AI-powered web applications, machine learning models, REST APIs, real-time web research systems, and data-driven applications. Skilled in Python development, data preprocessing, model evaluation, asynchronous web scraping, API integration, authentication, and database management. Passionate about solving real-world problems through scalable software and intelligent solutions.

INTERNSHIP EXPERIENCE

Cloud & DevOps Intern — Demy Software Solutions

Jun 2024 – Jul 2024

Visakhapatnam, India

- Learned fundamentals of AWS, Linux, Git, and DevOps practices.
- Gained knowledge of AWS EC2, CI/CD with Jenkins, and Docker containerization.
- Understood basic cloud deployment, automation, and version-control workflows.

Data Analytics Intern — APSSDC

May 2025 – Jun 2025

Andhra Pradesh State Skill Development Corporation · Online

- Completed a summer internship focused on Python programming, data analysis, and machine learning fundamentals.
- Built a Breast Cancer Detection project using Python, Pandas, NumPy, Matplotlib, and Scikit-learn.
- Developed and evaluated a machine learning classification model using Logistic Regression.
- Created visualizations to analyze the dataset and communicate model results effectively

PROJECTS

AI-Based Automated Social Media Content Creation Platform [Github](#)

Next.js (App Router), React, Node.js, Express, MongoDB, JWT, Bcrypt, OpenAI API, Meta Graph API, Cloudinary

- Built a full-stack platform to manage institutional events from content creation and approval to automated social media publishing.
- Designed a role-based workflow User → Draft → Admin → Published with secure JWT-based authentication and Bcrypt password hashing.
- Integrated OpenAI API for AI-powered content generation and Meta Graph API for automated Instagram and Facebook publishing with scheduled posts using node-cron.
- Developed a MongoDB-backed system to manage users, events, generated content, approval status, and publishing data.

Live Web Search & Research Engine [Github](#)

Python, FastAPI, Next.js, TypeScript, Tailwind CSS, SQLite

- Developed a real-time web research platform designed to search and analyze information from multiple online sources.
- Implemented asynchronous web scraping to retrieve and process information from live web pages.
- Designed a research workflow that aggregates information from different sources and generates structured research results.
- Structured research results to make source information, findings, and supporting evidence easier to analyze.

Breast Cancer Detection

Python, Pandas, Numpy, Matplotlib, Scikit-learn

- Developed a machine learning model to classify breast tumors as benign or malignant.
- Performed dataset preprocessing and feature analysis to prepare data for model training.
- Evaluated the model using accuracy, precision, recall, F1-score, confusion matrix, and classification report.
- Applied StandardScaler and train-test split for model preparation and evaluation.

EDUCATION

B.Tech, Computer Science & Engineering (AI & ML) — GPA 8.38/10

Oct 2022 – Apr 2026

Nadimpalli Satyanarayana Raju Institute of Technology, Visakhapatnam, AP

Board of Intermediate (MPC) — 91.3%

Jun 2020 – May 2022

Sri Chaitanya Jr College, Visakhapatnam, AP

Board of Secondary Education — 10.0 GPA

Jun 2019 – May 2020

Navya Vikas EM High School, Visakhapatnam, AP

TECHNICAL SKILLS

- **Languages:** Python, Java, C, TypeScript, JavaScript, HTML/CSS, SQL.
- **Frontend:** React, Next.js (App Router), Redux, Tailwind CSS.
- **Backend:** Node.js, Express.js, FastAPI, REST APIs, JWT/Bcrypt Auth.
- **Databases:** MongoDB, MySQL, SQLite.
- **AI&ML:** Pandas, NumPy, Scikit-learn, Matplotlib
- **Tools&APIs:** Git, GitHub, OpenAI API, Meta Graph API, Cloudinary.

CERTIFICATIONS

- Introduction to Machine Learning and AI.
- Overview of Building AI Apps on AWS.
- Bootstrap Training — Spoken Tutorial Project, IIT Bombay.
- Data Analytics — APSSDC Summer Online Internship Program

PUBLICATIONS

- **Published:** "Application of Artificial Intelligence in Education," International Journal of Innovative Research in Computer Science.
- **Published:** AI-Based Automated Social Media Content Creation and Posting System — Technical Research Publication.